What QUBO? Where do the atoms go?

One variable = one atom. A1 designs the loss; A3 makes Pasqal's E equation approximate it.

A1 — design the QUBO

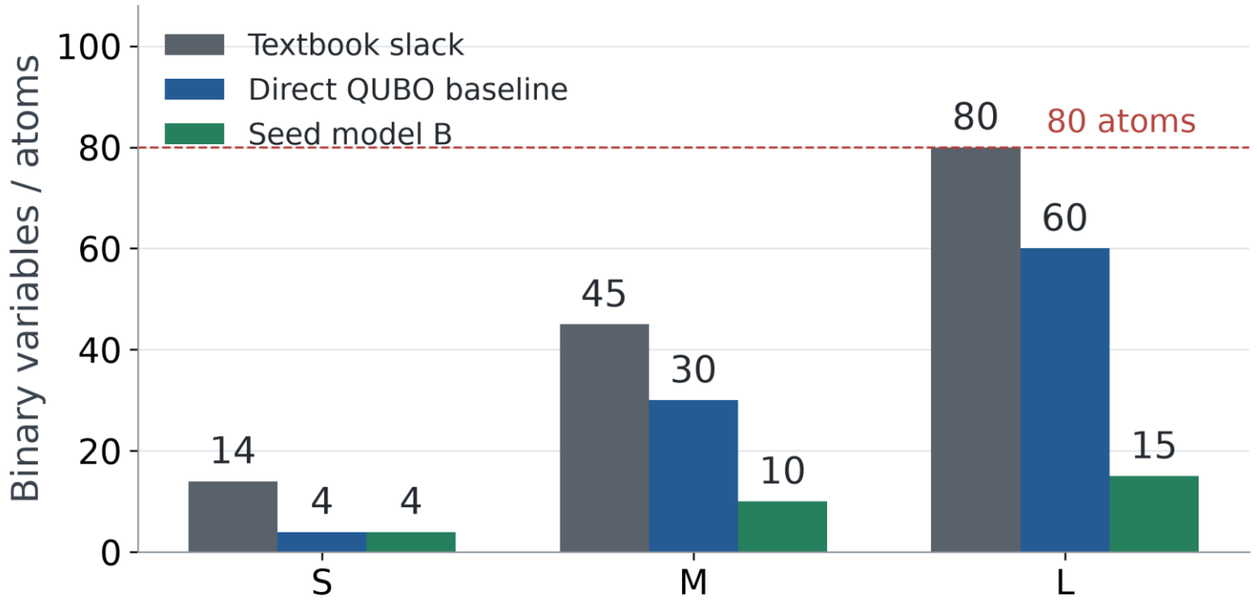
Cluster function: optimize.

Capacity: meet the passing line.

A3 — choose atom positions

Minimize disagreement between

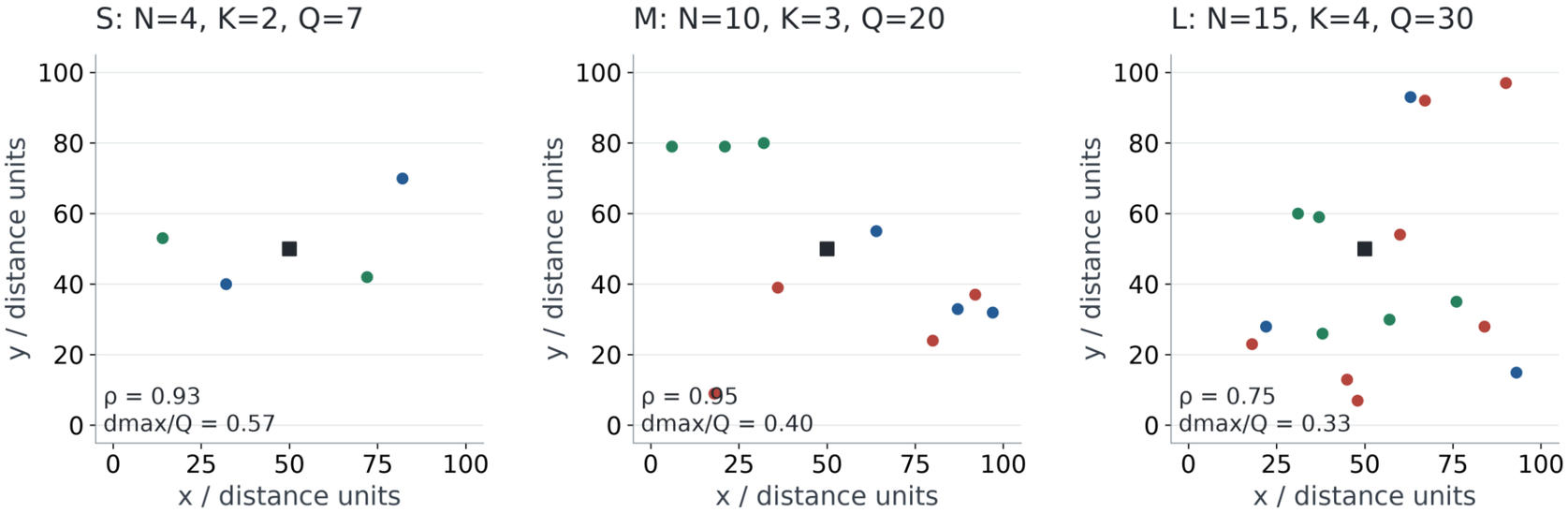
our QUBO and the machine's energy.



Workflow: A2 instances → A1 loss → A3 atom distances → A4 pulse. Model B is the working pipeline; the direct QUBO is the baseline.

Instances come with a feasible answer and a reference.

A2 is automated: seeded generation → certificate → validator → exact classical reference.



S_seed0 | C* = 110.14

Proven constrained clustering optimum

M_seed0 | C* = 353.04

Proven constrained clustering optimum

L_seed0 | C* = 475.36

Proven constrained clustering optimum

3 seeds per tier (0, 1, 2); maps show certificates, not EMU outputs. Depot fixed at (50, 50).

Cluster: optimize. Capacity: pass the line.

The desired constraint is $L_k \leq Q$. Our direct QUBO uses load balance as a slack-free surrogate.

$$\mathcal{L} = C_{\text{cluster}} + A\,P_{\text{assign}} + \lambda\,P_{\text{balance}}$$

CLUSTER FUNCTION — optimize

$$C = \sum_{k}\sum_{i<j} c_{ij}\,x_{ik}x_{jk}$$

CAPACITY — pass/fail: every load ≤ Q

$$P_{\text{balance}} = \sum_{k}(L_k - D/K)^2$$

Surrogate keeps rewarding balance.

$$\mathcal{L} = \text{const} + \sum_{a} a_a x_a + \sum_{a<b} b_{ab} x_a x_b$$

a_a = constant per atom | $b_{a\beta}$ = constant per atom pair | assignment: exactly one vehicle

M/L one-hot model shown; S uses one bit/customer. Balance still penalizes unequal feasible loads: it is not a flat pass/fail penalty.

Which QUBO terms fit Pasqal's E equation?

Same two slots: a constant per atom and an interaction per atom pair. Color identifies the inconsistency.

OUR QUBO (scaled)

$$s\mathcal{L} - \text{const} = \sum_a s a_a x_a + \sum_{a<b} s b_{ab} x_a x_b$$

PASQAL E EQUATION (Ω = 0)

$$E_{\text{hw}}/\hbar = -\delta \sum_a x_a + \sum_{a<b} C_6 \, r_{ab}^{-6} \, x_a x_b$$

per-atom constant

per-pair constant (set by distance)

We set: δ (laser) and r (positions).

We cannot set: C_6, the $1/r^6$ law.

MISMATCH: each atom wants its own a_a; the laser gives one −δ.

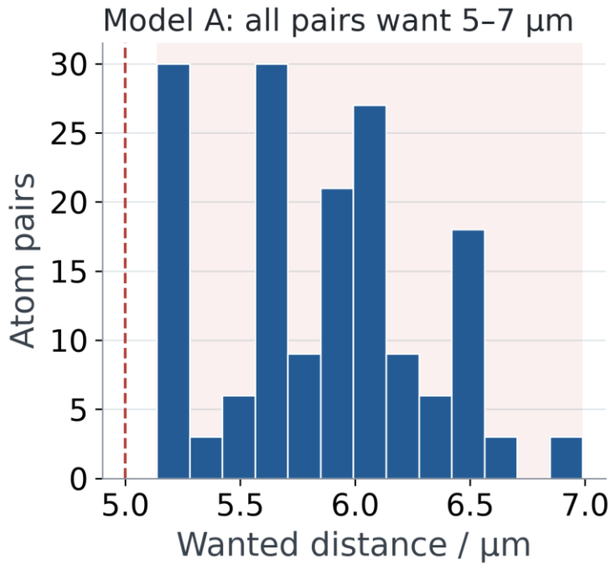
MATCH: both always positive.

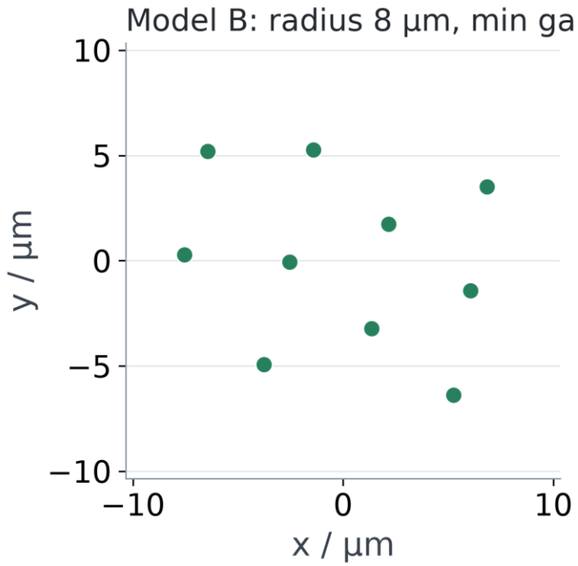
MISMATCH: some pairs want 0; each M atom wants 11 partners.

M/L: linear mismatch is constant on valid one-hot assignments, but changes invalid states. This does not hold for S.

Atom distance: the direct QUBO asks for an impossible layout.

Wanted distance $r_{a\beta} = (C_6 / U_{a\beta})^{1/6}$. Device: spacing ≥ 5 µm (red line), radius ≤ 38 µm.





Model A: measured limit

Too many pairs want the same short distance. M: true optimum ranks ≈10^4 of 59,049; η ≈ 1.

Model B: atoms = customer map

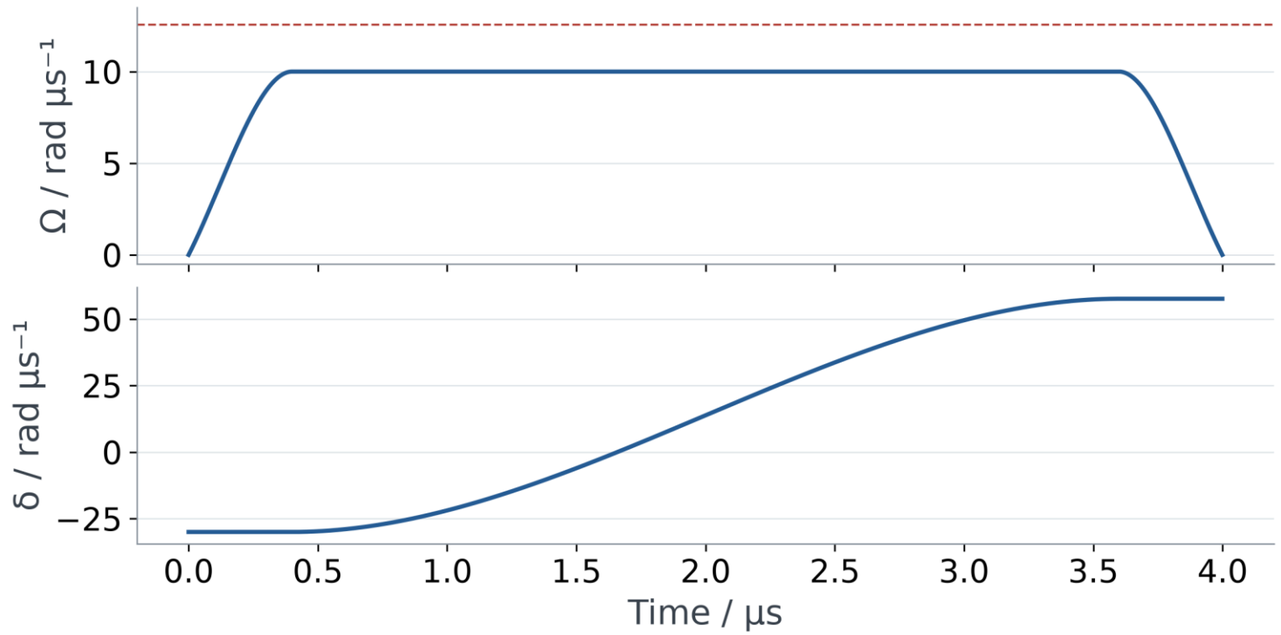
One atom per customer (4/10/15). Distance is the customer distance.

$$\mathbb{E}[C \mid F] = \frac{1}{P(F)} \sum_{x \in F} p(x \mid \mathbf{r}, \Omega, \delta, T)\, C(\mathrm{decode}(x))$$

M_seed0. Ranks and η are logged runs (PLAN §6). Expected loss is scored on feasible shots; P(F) is reported beside it.

Fix amplitude and duration; tune detuning per instance.

Proposed search policy, not completed optimization. All candidates must pass Pulser device validation.



Fixed within each tier — proposed

Ωpeak = 10 rad/µs; T = 4,000 ns

Starting policy shared across S/M/L.

Per-instance detuning grid

Direct: δf = {0.8, 1, 1.2} × δmap

Seeds: grid inside candidate K window

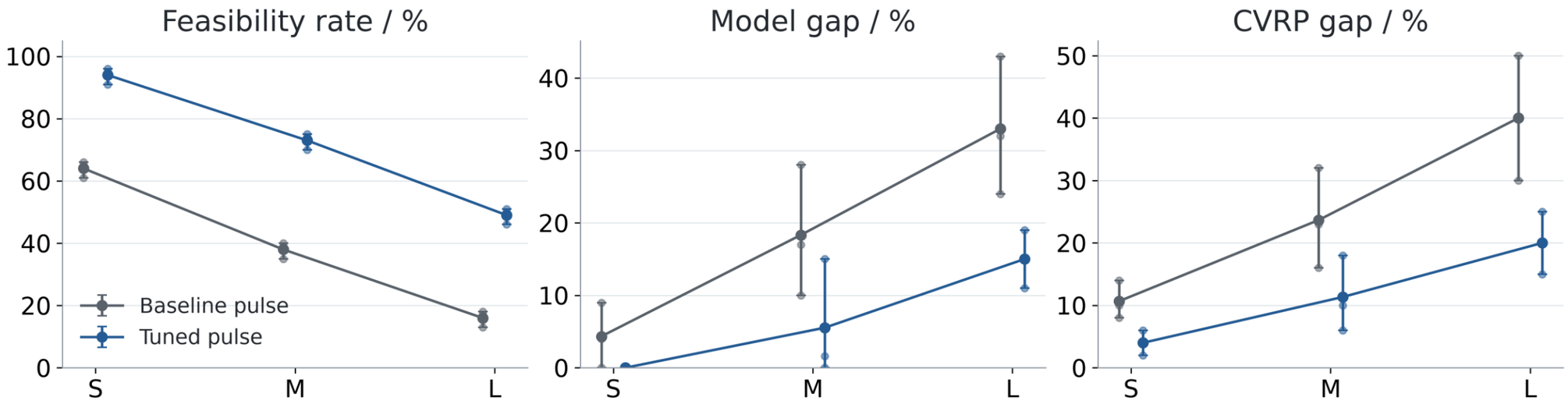
$\delta_0 = -\{0.5, 1, 1.5\} \times \Omega peak$

Check range and sequence legality first.

Shown: Pulser-constructed M_seed0 direct candidate, δf = −s mean(a). Seed-window cardinality needs a full-state check.

Report feasibility, model gap, CVRP gap and atoms.

MOCK performance until real EMU output exists. Keep direct QUBO and seed model B results separate.



Known direct-model limit: M_seed2

329.21 → 374.60 clustering cost (+13.8%).

Atoms A: 4 / 30 / 60; B: 4 / 10 / 15.

B needs random-seed baseline + decode audit.

MOCK: means and min–max across 3 seeds. CVRP gap uses exact classical routing. AI assisted code, figures and drafting.